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武汉大学 罗婷瑜

学科背景

专业背景

武汉大学地理信息科学（数字地图与空间信息工程）＋ 计算机科学与技术

总GPA **3.87/4**，均分**91.12**，大三学年综合素质测评专业**第一**

核心课程

地理信息系统原理

数据结构

空间分析

图形图像软件应用

地图设计与编制

空间信息可视化

空间数据库原理

计算机网络

地图艺术设计

面向对象的软件开发

数理基础好，专业课扎实

高等数学B1	高等数学B2	线性代数	概率论与数理统计	大学物理	数据统计分析SPSS
91	99	96	93	97	94
网络GIS与位置服务	地图投影与空间变换	数字摄影测量	地理信息系统实践	GIS+行业创新应用	地理信息科学研究进展
95	95	96	95	99	93

CPGIS2022论文发表

国际地理信息与技术大会（EI国际会议）

Comprehensive Evaluation of Emergency Shelters in Wuhan City Based on GIS（第一作者）

Comprehensive Evaluation of Emergency Shelters in Wuhan City Based on GIS

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***Abstract*—Emergency shelters, which reflect the city's ability to respond to and deal with major public emergencies to a certain extent, are essential to a modern urban emergency management system. This paper is based on spatial analysis methods, using Analytic Hierarchy Process to analyze the suitability of the 28 emergency shelters in Wuhan City. The Technique for Order Preference by Similarity to an Ideal Solution is further used to evaluate the accommodation capacity of emergency shelters in central urban areas, which provides a reference for the optimization of existing shelters and the site selection of new shelters, and provides a basis for improving the service capacity of shelters. The results show that the overall situation of emergency shelters in Wuhan is good, with 96% of the places reaching the medium level or above, but the suitability level needs to be further improved, especially the effectiveness and accessibility. Among the seven central urban areas in Wuhan, Hongshan District has the strongest accommodation capacity while Jianghan District has the weakest, with noticeable differences.**

***Keywords*—Wuhan City; Emergency shelter; Spatial analysis; Analytic Hierarchy Process**

I. Introduction

With the rapid development of cities and the continuous improvement of city governance systems, urban emergency management has become the focus of work. Emergency shelter is significant as it is a major component of city emergency resources and the important guarantee of people's life safety. Nowadays, emergency shelters are required to be distributed evenly, safe and reliable, adaptive to local conditions, useful both in normal times and in an emergency as well as planned in the long term. However, at present, the construction of large emergency shelters has the problems of insufficient scale, uneven distribution, and incomplete functions.

Many current evaluations of emergency shelters focus on site selection and optimization, planning and design, spatial pattern, and serviceability. For instance, Japanese scholars Tetsuya and Kayoko [1] use Geographic Information Systems(GIS) and apply statistical methods and public open data related to population and emergency shelters. The present research aims to conduct a suitability analysis for the emergency shelter allocation quantitatively. Gu et al. [2] evaluated the rationality of the spatial layout of emergency shelters in the central urban area of Heyuan City from four aspects: accessibility, service area ratio, per capita accessible refuge area ratio, and population allocation gap, but did not combine the four aspects and conduct an overall evaluation. Xiong et al. [3] used AHP to evaluate the disaster reduction capability of the earthquake emergency shelters in Chaoyang District, Beijing, but did not evaluate the service capacity of the emergency places. Yin et al. [4] used AHP to evaluate the site selection of emergency shelters in Tianjin, combined with GIS spatial analysis methods to evaluate the satisfaction of 14 shelters in the central urban area. However, they did not establish a complete satisfaction evaluation system and did not compare and analyze the service capacity of the emergency shelters in each district.

Based on GIS spatial analysis methods, in this paper, we use AHP and Technique for Order Preference by Similarity to an Ideal Solution(TOPSIS) methods to evaluate the site selection and accommodation capacity of emergency shelters. The factors considered are relatively complete, and the analysis of population data is also combined. We clearly and intuitively demonstrated the current construction's shortcomings, thus facilitating future planning and layout of emergency shelters.

II. Study Area and Data Sources

A. Overview of the study area

In this work, we study the emergency shelters in Wuhan City, Hubei Province, an important scientific and educational base, industrial base, and comprehensive transportation hub. It is located at the intersection of the golden waterway of the Yangtze River and the main artery of the Beijing-Guangzhou Railway and is known as the Nine Provinces Passage. Wuhan is also in an advantageous position in the economic geography circle of China and is the core city of the Yangtze River Economic Belt. Wuhan has a total area of 8,569 square kilometers, with a permanent population of about 11.45 million in 2021. Up to now, Wuhan has jurisdiction over 13 districts, including Jiang'an District, Jianghan District, Qiaokou District, Hanyang District, Wuchang District, Qingshan District, Hongshan District, Caidian District, Jiangxia District, Huangpi District, Xinzhou District, Dongxihu District, and Hannan District.

B. Data sources

The data of Wuhan's administrative divisions, traffic network, water system, etc., are from OpenStreetMap[1] and were pretreated through ArcGIS software. The directory and relevant information of Wuhan emergency shelters, major hazard sources, hospitals, fire stations, and public security organs originate from Wuhan public data open platform[2]. The coordinates of these places are obtained through the Baidu map coordinate picking system and converted to the earth coordinates. The DEM data of Wuhan comes from geospatial data cloud[3]. The data on Wuhan City area, resident population, population density, and other data comes from the Wuhan City Statistical Yearbook (2021)[4].

[1] https://www.openhistoricalmap.org

[2] https://data.wuhan.gov.cn

[3] https://www.gscloud.cn

[4] http://tjj.wuhan.gov.cn/tjfw/tjnj/

III. Suitability Evaluation of Emergency Shelters

A. Evaluation system

Based on the functions and significance of emergency shelter, the suitability evaluation of emergency shelter is decomposed into target layer, criterion layer, and index layer by AHP. Combined with Construction Standard of Urban Community Emergency Shelters (JB 180-2017) and Wuhan Emergency Project Management Measures, the evaluation indexes are selected from the four dimensions of effectiveness, safety, reachability and supportability, and finally the analytic hierarchy process model for the suitability evaluation of emergency shelters in Wuhan is constructed (see Fig. 1).

B. Calculation of the model

In this study, following the working methods of Chu et al. [5], the relative importance of each element are obtained and then the judgment matrices of each criterion layer and index layer are constructed, and the weights are calculated after passing the consistency test (shown in Table 1).

TABLE I
Each evaluation index and its weight value

Index Layer	Weight(W)
Refuge area C_1	0.0930
Capacity C_2	0.0930
Functional facilities C_3	0.0465
Topography C_4	0.0841
Distance from major hazard installations C_5	0.1780
Avoid geological hazard-prone areas C_6	0.0519
Avoid hidden points of hydrology C_7	0.0519
Distance from nearest hospital C_8	0.0525
Distance from nearest fire station C_9	0.0597
Distance from nearest public security unit C_{10}	0.0294
Road accessibility C_{11}	0.1362
Daily management C_{12}	0.0928
Sustainable development C_{13}	0.0510

C. Calculation of the comprehensive index

Calculate the comprehensive indexes of emergency shelters in Wuhan, and get the results of suitability evaluation according to the comprehensive index. The calculation process mainly includes:

1) Establish a geographic information database of emergency shelters in Wuhan City using ArcGIS.
2) Give a score to each index. C_3, C_{11} and C_{13} were artificially graded according to the data of each site, and were given 2, 5 and 8 points respectively. Other indexes are divided into 10 levels by using natural breakpoint method in ArcMap, and 1-10 points are given according to the levels to minimize the influence of subjective factors.
3) Use the following formula to calculate the comprehensive index of each emergency shelter.

$$P = \sum_{i=1}^{13} W_i F_i, i = 1, 2, ..., 13$$

4) Conduct suitability evaluation. Divided the calculated comprehensive indexes into five grades (Table 2). The suitability evaluation results of emergency shelters in Wuhan are shown in Table 3 and Figure 2.

TABLE II
Classification of suitability grades

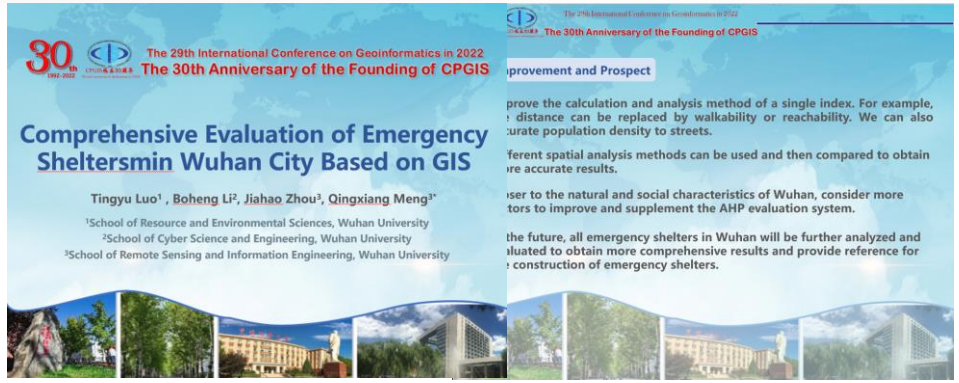
Composite Index(P)	Grade(G)	Grade Description
8.001-10.000	A	Better suitability, it is reasonable in terms of effectiveness, safety, reachability and supportability. It is a good emergency shelter and needs to be maintained and improved.
6.001-8.000	B	Good suitability, it needs to maintain good aspects, and make targeted improvements of deficiencies.
4.001-6.000	C	Medium suitability, it basically meets the functional requirements of emergency shelter and needs further improvement.
2.001-4.000	D	Poor suitability, it still has great room for improvement.
0.000-2.000	E	Poorer, it is not suitable to be used as an emergency shelter.

D. Results analysis

Overall, the emergency shelters in Wuhan are in good condition. Among the 28 selected emergency shelters, 14 are suitable, 13 are medium, and one is sub-standard. Qingshan Park, located in Qingshan District, has the highest suitability level, while the Zhuyehai Park in Qiaokou District is the lowest.

From the effectiveness perspective, more than 57 percent of emergency shelters scored less than 5 points, indicating that they have limited ability to provide help to surrounding citizens in response to emergencies, which is mainly limited by the number of people they can accommodate. Baibuting Garden, located in Jiang'an District, needs to be strengthened because of its small area, the small number of people, and low level of effectiveness.

In terms of safety, the overall situation is quite good. The emergency evacuation places of Lianzheng Park, International Convention and Exhibition Center, Zhuyehai Park, and Wenti Square have potential safety hazards because they are close to



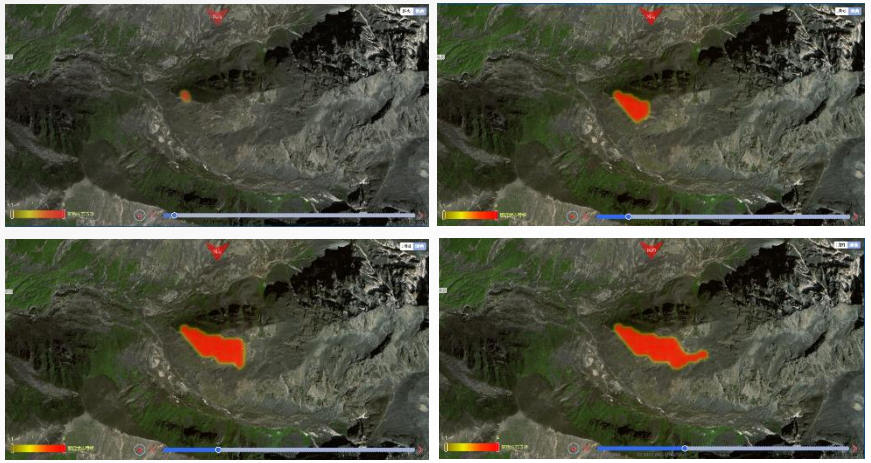
大致内容

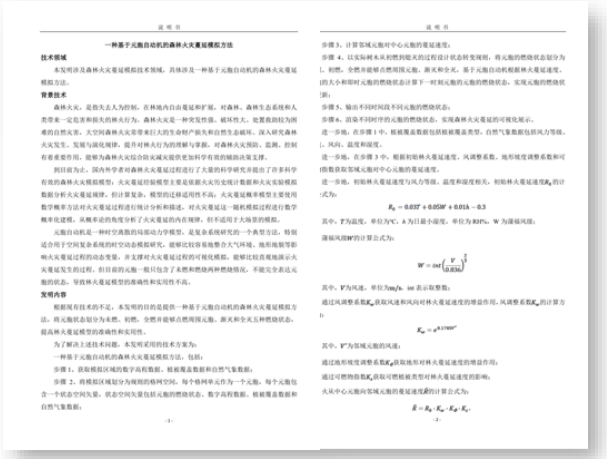
采用 AHP 模型对选取的28个武汉市应急避难场所进行适宜性分析

采用 TOPSIS 模型对武汉市中心城区应急避难场所的容纳能力进行评价

发明专利申请

一种基于元胞自动机的森火蔓延模拟方法（第一发明人）





说 明 书

一种基于元胞自动机的森林火灾蔓延模拟方法

技术领域

本发明涉及森林火灾蔓延模拟技术领域，具体涉及一种基于元胞自动机的森林火灾蔓延模拟方法。

背景技术

森林火灾，是指失去人为控制，在林地内自由蔓延和扩展，对森林、森林生态系统和人类带来一定危害和损失的林火行为。森林火灾是一种突发性强、破坏性大、处置救助较为困难的自然灾害，大空间森林火灾常常来巨大的生命财产损失和自然生态破坏。深入研究森林火灾发生、发展与演化规律，提升对林火行为的理解与掌握，对森林火灾预防、监测、控制有着重要作用，能够为森林火灾综合防灾减灾提供更加科学有效的辅助决策支撑。

到目前为止，国内外学者对森林火灾蔓延过程进行了大量的科学研究并提出了许多科学有效的森林火灾模拟模型，火灾蔓延经验模型主要是根据火灾历史统计数据和火灾实验模拟数据分析火灾蔓延规律，经计算复杂，模型的迁移适用性不高。火灾蔓延概率模型主要使用数学概率方法对火灾蔓延过程进行统计分析和推进，对火灾蔓延这一随机模拟过程进行数学概率化建模，从概率论的角度分析了火灾蔓延的内在规律，但不适用于大场景的模拟。

元胞自动机是一种时空离散的局部动力学模型，是复杂系统研究的一个典型方法，特别适合用于空间复杂系统的时空动态模拟研究，能够比较容易地整合大气环境、地形地貌等影响火灾蔓延过程的动态变量，并支撑对火灾蔓延过程的可视化模拟，能够比较直观地展示火灾蔓延发生的过程，但目前的元胞一般只包含了未燃和燃烧两种燃烧情况，不能完全表达元胞的状态，导致林火蔓延模型的准确性和实用性不高。

发明内容

根据现有技术的不足，本发明的目的是提供一种基于元胞自动机的森林火灾蔓延模拟方法，将元胞状态划分为未燃、初燃、全燃并能够点燃周围元胞，源火和全火五种燃烧状态，提高林火蔓延模型的准确性和实用性。

为了解决上述技术问题，本发明采用的技术方案为：

一种基于元胞自动机的森林火灾蔓延模拟方法，包括：

步骤 1，获取模拟区域的数字高程数据，植被覆盖数据和自然气象数据；

步骤 2，将模拟区域划分为规则的格网空间，每个格网单元作为一个元胞，每个元胞包含一个状态空间矢量，状态空间矢量包括元胞的燃烧状态，数字高程数据，植被覆盖数据和自然气象数据；

-1-

说 明 书

步骤 3，计算邻域元胞对中心元胞的蔓延速度；

步骤 4，以实际树木从初燃到熄灭的过程设计状态转变规则，将元胞的燃烧状态划分为 I，初燃，全燃并能够点燃周围元胞，源火和全火，基于元胞自动机根据林火蔓延速度，[illegible]的大小和即时元胞的燃烧状态计算下一时刻元胞的元胞的燃烧状态，实现元胞的燃烧状[illegible]；

步骤 5，输出不同时间段不同元胞的燃烧状态；

步骤 6，渲染不同时序的元胞的燃烧状态，实现森林火灾蔓延的可视化展示。

进一步地，在步骤 1 中，根据覆盖数据包括植被覆盖类型，自然气象数据包括风力等级、[illegible]、风向、温度和湿度。

进一步地，在步骤 3 中，根据初始林火蔓延速度、风调整系数、地形坡度调整系数和可燃物数获取邻域元胞对中心元胞的蔓延速度。

进一步地，初始林火蔓延速度与风力等级、温度和湿度相关，初始林火蔓延速度R_0的计算公式为：

$$R_0 = 0.03T + 0.05W + 0.01h - 0.3$$

其中，T为温度，单位为℃，h为日最小湿度，单位为 RH%，W 为蒲福风级；

蒲福风级W的计算公式为：

$$W = int\left(\frac{V}{0.836}\right)^{\frac{2}{3}}$$

其中，V为风速，单位为m/s，int 表示取整数；

通过风调整系数K_w获取风速和风向对林火蔓延速度的增益作用，风调整系数K_w的计算为：

$$K_w = e^{0.1166V'}$$

其中，V'为邻域元胞的风速；

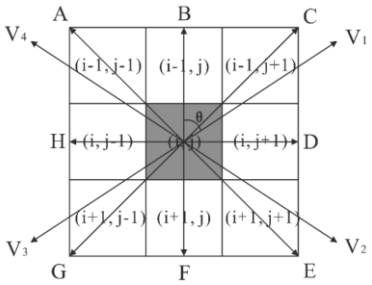
通过地形坡度调整系数K_φ获取地形对林火蔓延速度的增益作用；

通过可燃物指数K_s获取可燃植被类型对林火蔓延速度的影响；

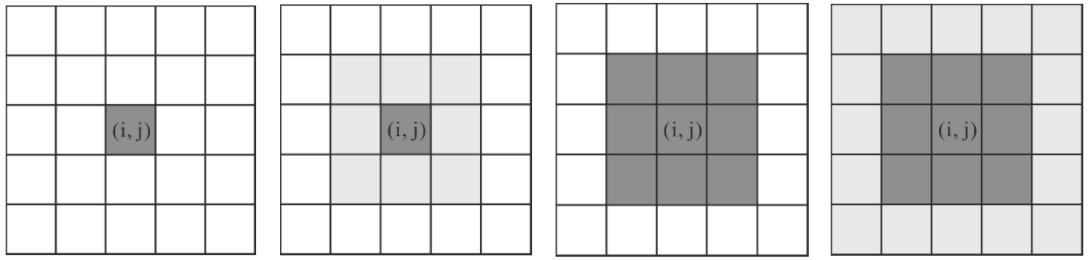
火从中心元胞向邻域元胞的蔓延速度$\hat{R}$的计算公式为：

$$\hat{R} = R_0 \cdot K_w \cdot K_\varphi \cdot K_s.$$

-2-



元胞八邻域示意图



元胞自动机模拟林火蔓延示意图

主要思路

获取高程数据、植被覆盖数据、自然气象数据

将划分为规则格网，每个格网单元作为一个元胞

计算中心元胞向邻域元胞的蔓延速度

$$R_0 = 0.03T + 0.05W + 0.01h - 0.3$$

$$R = R_0 \cdot K_w \cdot K_\varphi \cdot K_s$$

设计**状态转变规则**，计算下一时刻元胞状态，实现元胞状态的更新

发明专利申请

核心改进点一

风、地形调整系数 $K_w = e^{0.1783V}$ $K_\varphi = e^{0.355(\tan\varphi)^{1.2}}$

考虑风向和元胞邻域

$$\tilde{R} = R_0 K_s \begin{bmatrix} (K_w K_\varphi)_{i-1,j-1} \\ (K_w K_\varphi)_{i-1,j} \\ (K_w K_\varphi)_{i-1,j+1} \\ (K_w K_\varphi)_{i,j+1} \\ (K_w K_\varphi)_{i+1,j+1} \\ (K_w K_\varphi)_{i+1,j} \\ (K_w K_\varphi)_{i+1,j-1} \\ (K_w K_\varphi)_{i,j-1} \end{bmatrix}。$$

$$R_{i-1,j-1} = R_0 \cdot K_S \cdot e^{0.1783V\cos(315^\circ-\theta)} \cdot e^{3.533(-1)^G\left|\frac{h_{k,l}-h_{i,j}}{\sqrt{2}a}\right|^{1.2}}$$

$$R_{i-1,j} = R_0 \cdot K_S \cdot e^{0.1783V\cos\theta} \cdot e^{3.533(-1)^G\left|\frac{h_{k,l}-h_{i,j}}{a}\right|^{1.2}}$$

$$R_{i-1,j+1} = R_0 \cdot K_S \cdot e^{0.1783V\cos(45^\circ-\theta)} \cdot e^{3.533(-1)^G\left|\frac{h_{k,l}-h_{i,j}}{\sqrt{2}a}\right|^{1.2}}$$

$$R_{i,j+1} = R_0 \cdot K_S \cdot e^{0.1783V\cos(90^\circ-\theta)} \cdot e^{3.533(-1)^G\left|\frac{h_{k,l}-h_{i,j}}{a}\right|^{1.2}}$$

$$R_{i+1,j+1} = R_0 \cdot K_S \cdot e^{0.1783V\cos(135^\circ-\theta)} \cdot e^{3.533(-1)^G\left|\frac{h_{k,l}-h_{i,j}}{\sqrt{2}a}\right|^{1.2}}$$

$$R_{i+1,j} = R_0 \cdot K_S \cdot e^{0.1783V\cos(180^\circ-\theta)} \cdot e^{3.533(-1)^G\left|\frac{h_{k,l}-h_{i,j}}{a}\right|^{1.2}}$$

$$R_{i+1,j-1} = R_0 \cdot K_S \cdot e^{0.1783V\cos(225^\circ-\theta)} \cdot e^{3.533(-1)^G\left|\frac{h_{k,l}-h_{i,j}}{\sqrt{2}a}\right|^{1.2}}$$

$$R_{i,j-1} = R_0 \cdot K_S \cdot e^{0.1783V\cos(270^\circ-\theta)} \cdot e^{3.533(-1)^G\left|\frac{h_{k,l}-h_{i,j}}{a}\right|^{1.2}}$$

$R_0 K_s (K_w K_\varphi)_{k,l}$ 表示火从元胞（i，j）向元胞（k，l）蔓延的速度分量

核心改进点二

元胞状态的划分：

将森林元胞的状态划分为：S=0未燃，S=1初燃，S=2全燃并具有点燃周围元胞的能力，S=3渐灭，S=4全灭

遍历火场中燃烧状态可能改变的元胞，即满足：

$$\left(S = 0 \,\&\&\, \exists S_{邻} = 2\right) || 0 < S < 4$$

元胞状态转变规则：

若此时 S=0，可燃且邻域存在状态为 S=1 的初燃元胞，则计算得到下一时刻该元胞值；

若此时 S=1 恰好初燃，则下一时刻 S=2，全燃；

若此时 S=2 且邻域 S≥2 或不可燃，则 S=3，逐渐熄灭；

若此时 S=3，则下一时刻 S=4，完全熄灭。

开发能力

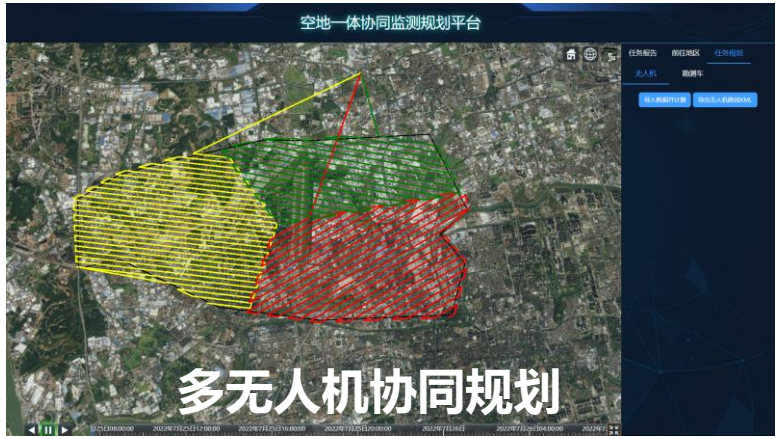
重大危险源态势感知

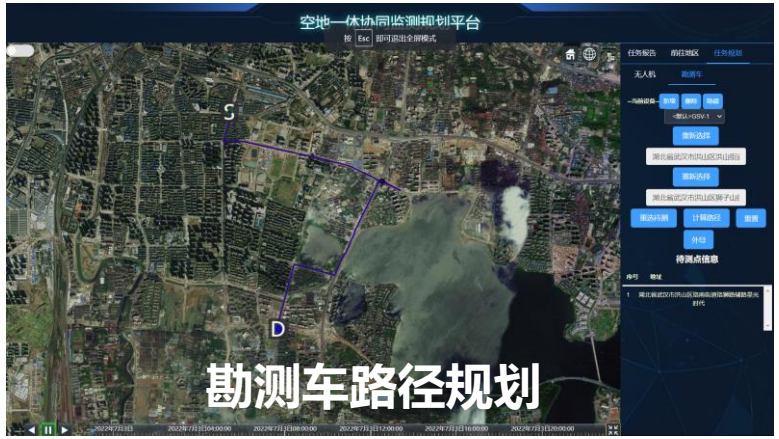
人防工程缓冲区态势感知

应急资源的分析与评价

应急资源可视化平台的开发

空天地一体监测平台的开发

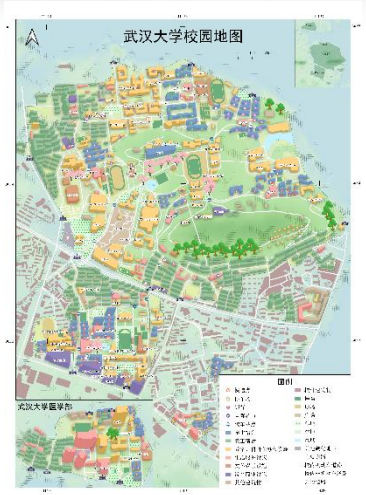


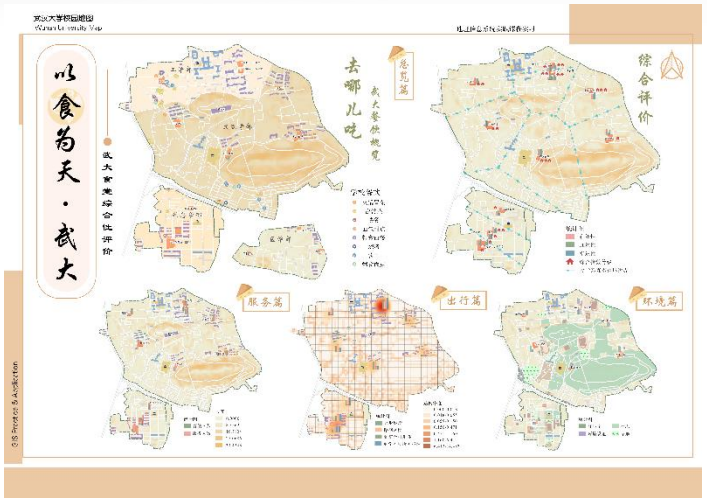


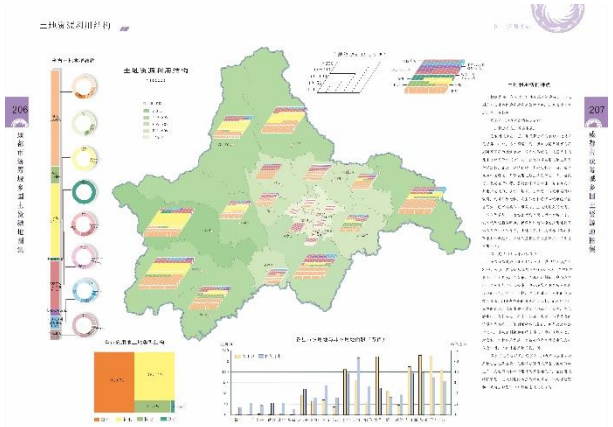
智慧城市与应急管理

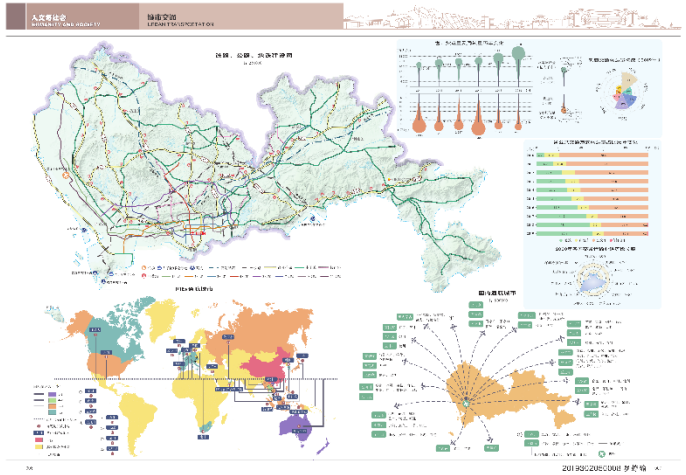
应急避难场所可视化

地图设计制作能力









“兹游奇绝处，烟雨任平生”

地理与历史的结合，**文化地图**编制的探索

于课程作业中提高地图设计、专题图编制能力

空间分析能力

地区社会弱势性空间格局分析

区域醉酒空间格局及其影响因素分析

社区生活圈可步性测度

公园绿地可达性及其优化选址

区域多中心性测度

耕地破碎化空间格局及其影响因素分析

校园食堂分布适宜性评价与选址规划（右图）

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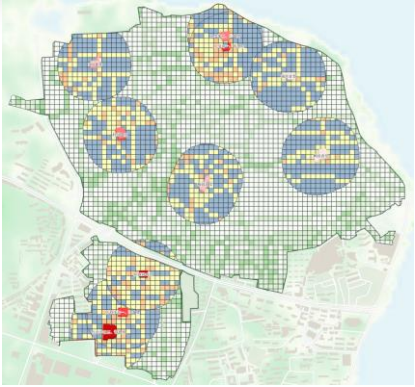
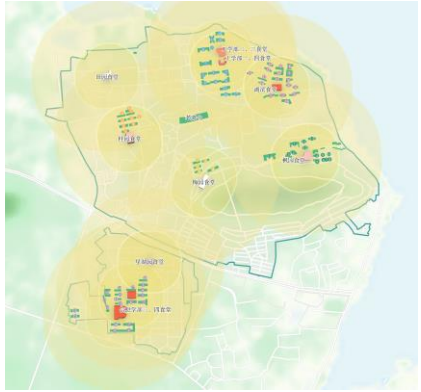
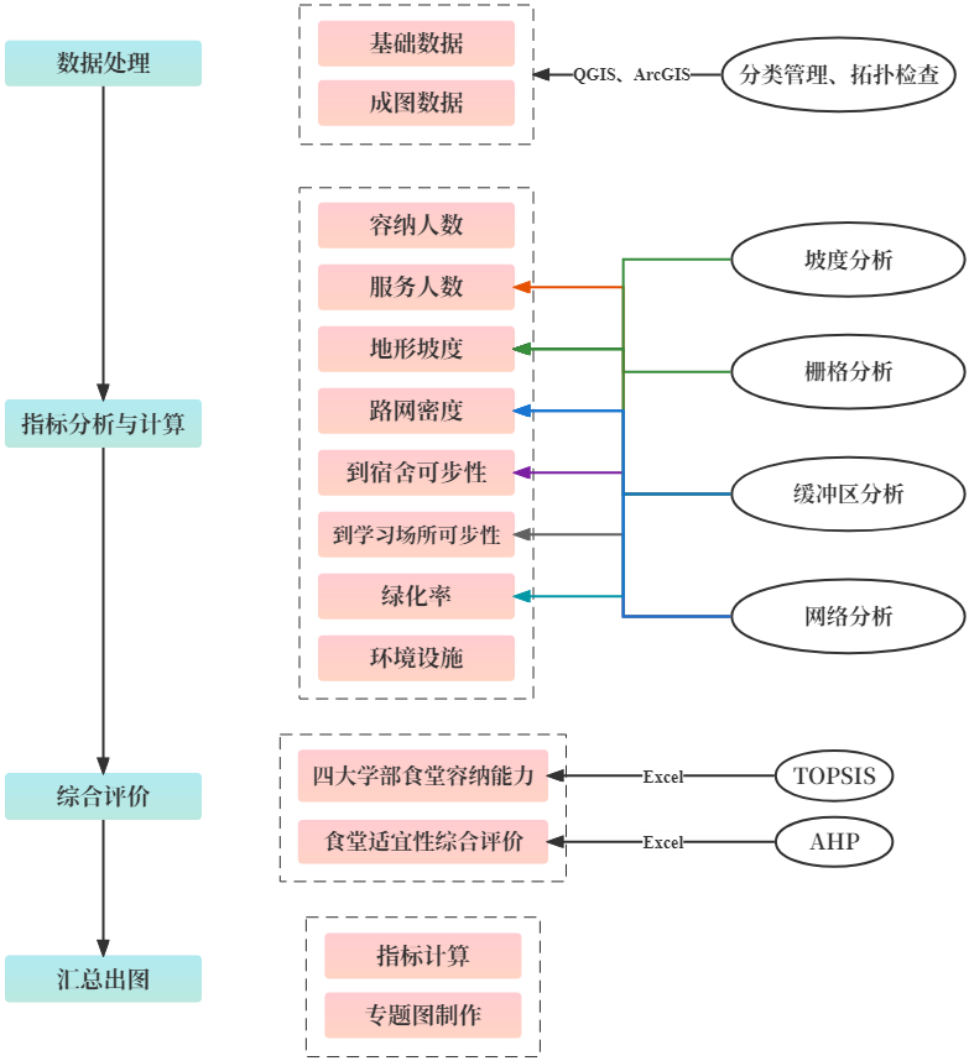
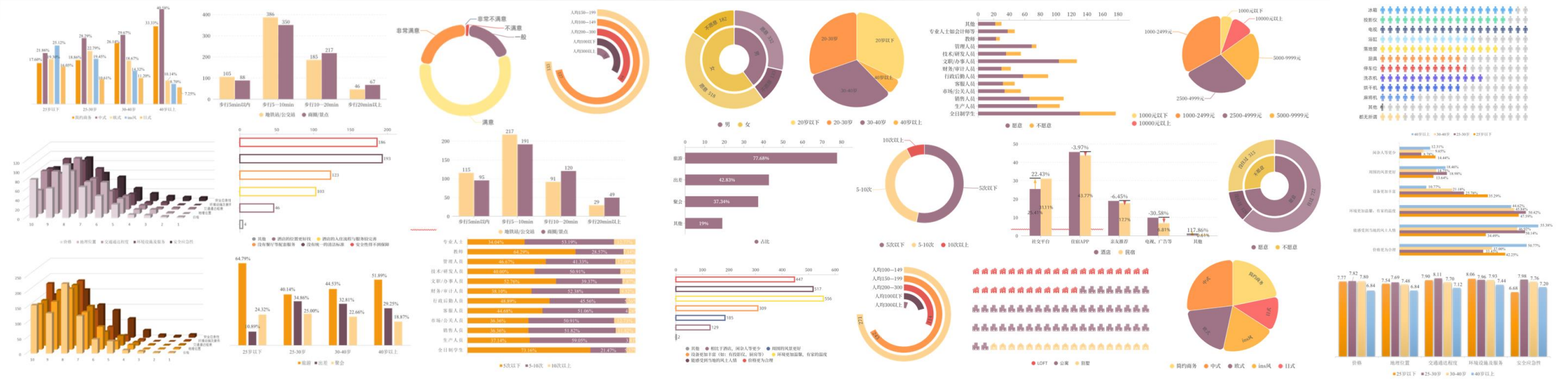






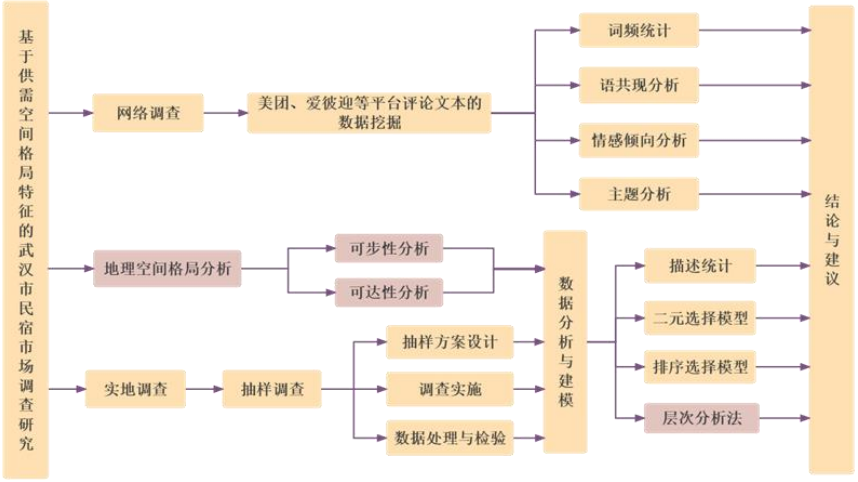
学科交叉+数据分析



何曾住处限西东

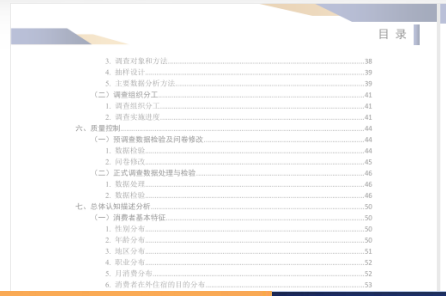
——基于供需空间格局特征的武汉市民宿市场调查研究

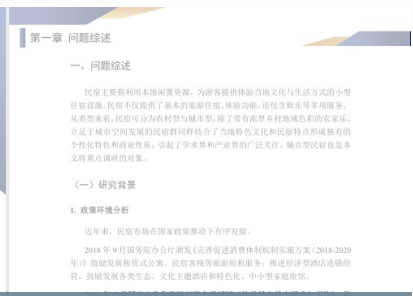
面向民宿市场供需平衡发展的目标，采取网络爬虫和问卷调查相结合的方法，利用GIS方法识别武汉市民宿市场空间格局特征，挖掘民宿市场发展背后隐藏的价值信息，提出民宿市场发展的优化策略。

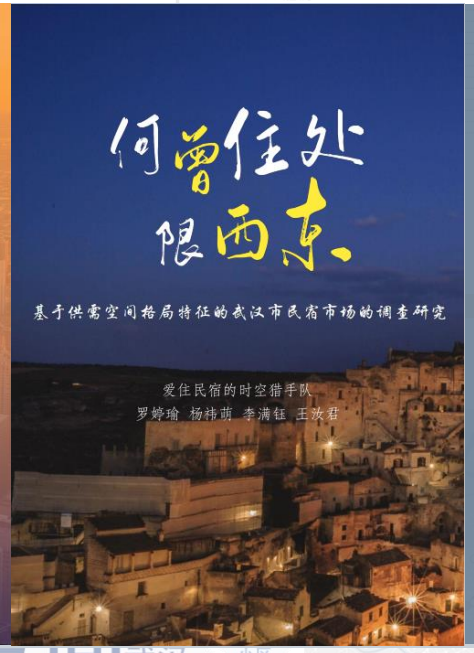


“GIS+” 一次**学科交叉**的探索

学科交叉+数据分析







整体思路+进度规划

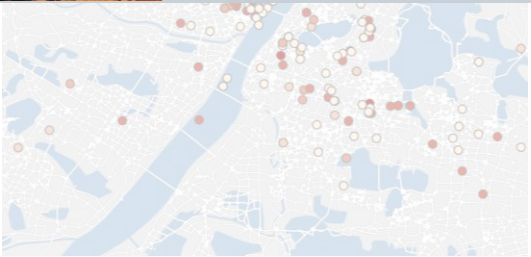
评论文本数据的挖掘与分析

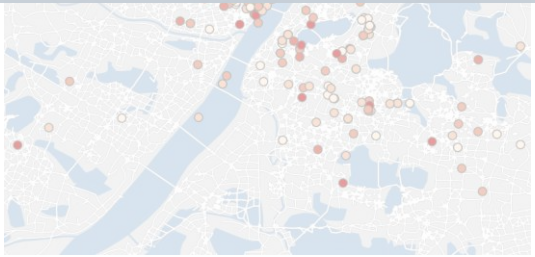
问卷数据的挖掘与分析

排序选择模型的运用

AHP模型的结合

利用SPSS、R、Python等分析处理数据





创新思维训练

武汉城市圈开放数据应用大赛二等奖
2022.01

市场调查与分析大赛湖北赛区三等奖
2022.5

“挑战杯”创业计划大赛湖北省**金奖**
2022.6

- 全国大学生测绘学科创新创业智能大赛 **国家特等奖**
- 互联网+大学生创新创业大赛湖北省**金奖**

2022.7

互联网+大学生创新创业大赛**全国总决赛金奖**
2022.10

- “挑战杯”创业计划大赛全国总决赛铜奖
- 雄安新区“科技•人才•创新周”交流学习

2023.3

……

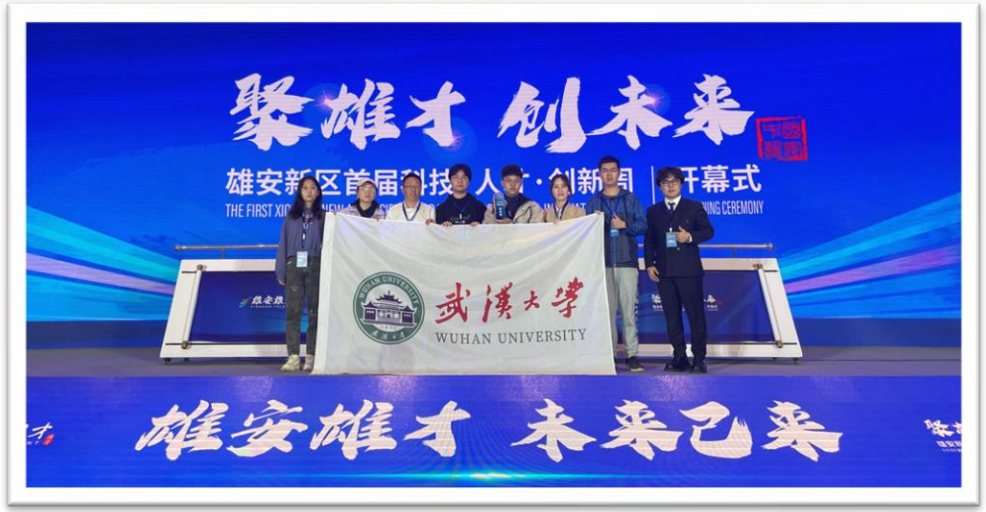
作为团队核心成员（3/15）参加创新创业比赛，在其中不断培养互联网创新思维，锻炼发现并解决社会中的实际问题的能力

未来目标

从自身学科与专业出发 → 交叉融合，实践探索

“赛”为重 → “创”为重，不断创新

学习交流



参加雄安新区首届科技•人才•创新周活动，与清北等高校优秀团队交流学习

THANK YOU

武汉大学 罗婷瑜